Local Distribution

LDC City Gate to End-Use

Design Basis Accident — Military Action

DBA-GAS-FC4 — Facility Class Document

Issued Under: DBA-GAS Sector Framework

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Preface

Every other facility class in the DBA-GAS series is infrastructure that most people never encounter. Transmission pipelines run underground across hundreds of miles. Processing plants operate in remote producing regions. Storage fields are invisible geological formations. The local distribution system is different: it ends at the kitchen stove, the furnace thermostat, and the water heater in the basement. When it fails, the consequence is not a supply disruption felt in market prices or generation statistics. It is a cold house in January, a family without hot water, an older adult in a poorly insulated apartment in a city that has lost its gas supply and has no immediate alternative.

This document establishes the Design Basis Accident scenarios for deliberate military action against local natural gas distribution infrastructure. The facility class encompasses the city gate stations at which local distribution companies (LDCs) receive gas from the interstate transmission system, the district regulator stations that reduce pressure for neighborhood-level delivery, the distribution mains and service lines that carry gas to individual customers, and the SCADA and operational technology systems that manage and monitor the distribution network. It is the last mile of the natural gas supply chain — and uniquely among all the facility classes in the DBA-GAS series, its failure produces direct, immediate life-safety consequences for the residential population it serves.

The governing analytical distinction of FC4 is the re-pressurization constraint. When a transmission pipeline segment goes offline, restarting the compressor stations re-pressurizes it. When a distribution system loses pressure — fully depressurizes — no equivalent shortcut exists. Every service connection in the affected area must be physically visited by a qualified technician who verifies the customer's appliances are off, relights pilots, and confirms safe operation before gas pressure is restored to that customer. For a large urban distribution system serving hundreds of thousands of customers, this process takes weeks. There are not enough technicians. There is no way to accelerate it without creating gas explosion risks in residential buildings whose occupants may have left appliances open or whose pilot lights may have been extinguished by the pressure loss. The re-pressurization constraint is the governing recovery metric for every FC4 scenario — and it has no analog anywhere else in the DBA-GAS series.

The evidentiary anchors for FC4 are operational rather than catastrophic: the pattern of urban distribution outages following natural disasters — the 2018 Merrimack Valley gas explosions in Massachusetts, where over-pressurization of a distribution system caused dozens of fires and explosions across three communities; the extended distribution outages following Hurricane Sandy in 2012; and the systematic curtailment of distribution customers during the 2021 Winter Storm Uri transmission crisis. The DBA-MA extension asks what the Merrimack Valley scenario looks like when the initiating condition is deliberate adversarial action rather than an operational error — and what the Uri curtailment looks like when the transmission disruption is compounded by a simultaneous attack on the distribution infrastructure that would otherwise serve the customers who remain on gas.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the local natural gas distribution facility class. Its purpose is to define the bounding threat event that distribution system security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A distribution system security program designed against conventional physical threats will fail when a sophisticated adversary combines SCADA-based pressure management manipulation — driving a distribution district to zero pressure without triggering the automatic low-pressure shutoffs that would prompt an emergency response — with selective kinetic action against city gate stations, producing a systematic depressurization that leaves the utility facing the full re-pressurization constraint across a major urban service territory. The DBA-MA approach must account for the cyber-only pathway, the kinetic pathway, and the compound pathway where transmission and distribution infrastructure are attacked simultaneously.

1.2 Scope and Applicability

This document addresses local natural gas distribution facilities: the infrastructure through which local distribution companies receive gas from interstate or intrastate transmission pipelines at city gate stations and deliver it to residential, commercial, and small industrial customers through lower-pressure distribution systems. The facility class encompasses city gate stations, district regulator stations, distribution mains and service lines, measurement and regulation infrastructure, and the SCADA and operational technology systems that govern distribution system operations. LNG peaking facilities — small-scale liquefaction and storage systems operated by LDCs as emergency supply supplements — are within the scope of this document as part of the LDC's supply portfolio.

Interstate natural gas transmission pipelines and compressor stations are addressed in DBA-GAS-FC1. Gas processing plants are addressed in DBA-GAS-FC2. Underground storage fields are addressed in DBA-GAS-FC3. Large-scale LNG export and import terminals are addressed in DBA-GAS-UC-LNG. Where distribution system events produce consequences that propagate back into the transmission system — particularly where large distribution loads are suddenly curtailed, creating a transmission system pressure surge — this document addresses those consequences as they originate at the distribution interface without displacing the DBA-GAS-

FC1 analysis. The residential heating dependency identified as cross-sector consequence XC-4 in DBA-ES-GC-FC4 is addressed here from the gas sector's perspective, completing the cross-reference established in the electric sector series.

1.3 Relationship to Governing Documents

This document is issued under the DBA-GAS Sector Framework (v1.5), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the distribution-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories are inherited with distribution-specific adaptation. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the distribution SCADA and pressure management environment in Section 3.3. The residential heating dependency cross-referenced in DBA-ES-GC-FC4 (XC-4) and in DBA-EN-SF (OI-4, now addressed) is resolved by this document from the gas sector's analytical perspective.

1.4 The Last Mile — Direct Life-Safety and the Re-Pressurization Problem

Two characteristics of the local distribution facility class distinguish it from every other facility class in the DBA-GAS series and govern the entire analytical structure of this document.

The first is the directness of the life-safety consequence. Every other facility class in the DBA-GAS series produces supply disruptions whose life-safety consequence is indirect — mediated by market responses, alternative fuel options, and the time available for adaptation. A distribution system outage in a dense urban area during winter heating season produces direct life-safety risk for vulnerable residents — elderly, medically dependent, and economically isolated populations who cannot quickly find alternative heat sources and who may suffer hypothermia-related injury or death within days of losing gas service. This consequence is not proportional to the distribution system's engineering complexity or its position in the supply chain; it is disproportionately severe precisely because the distribution system serves the most vulnerable end of the demand curve.

The second is the re-pressurization constraint. A distribution system that has lost all gas pressure — whether through a city gate station failure, a major main rupture, or a systematic SCADA-induced depressurization — cannot be restored to service by a simple re-pressurization of the inlet. Every customer service connection must be physically inspected and re-lit before that customer's service can be restored. The reason is fundamental gas safety: when a distribution system loses pressure, gas appliances at every affected customer location may have been left in an unsafe state — pilot lights extinguished, gas valves partially open, and appliances whose automatic safety features rely on a continuous pilot flame. Restoring pressure to such a system without visiting every connection creates the conditions for a Merrimack Valley-type cascade of residential fires and explosions across the affected service territory.

For a large urban LDC serving 500,000 customers, the re-pressurization process requires deployment of every available field technician — and typically mutual aid technicians from other utilities — for a sustained operation lasting two to four weeks. The technician workforce is finite; no reserve pool can be rapidly mobilized. This is the governing recovery constraint for every Scenario A event in FC4, and it has no analog in the compressor replacement timelines, cold box procurement timelines, or blowout control contractor constraints that govern recovery in the other DBA-GAS facility classes.

2. Distribution System Profile and Asset Taxonomy

2.1 City Gate Stations

City gate stations are the interface between the interstate or intrastate transmission system and the LDC's distribution system. They perform three functions: pressure reduction from transmission system pressure (typically 200 to 1,200 psi) to distribution system operating pressure (typically 0.25 to 100 psi depending on the distribution tier); odorization — the addition of mercaptan odorant that makes odorless natural gas detectable by smell; and measurement of gas volumes received for custody transfer and billing. City gate stations range from simple single-regulator installations serving small communities to complex multi-train facilities with redundant regulators, filter separators, heating systems, and telemetry serving major metropolitan areas.

The city gate station is the highest-consequence single physical attack target in the distribution system. Its destruction eliminates all gas supply to the distribution system it feeds, regardless of the physical condition of the downstream distribution infrastructure. A major metropolitan LDC typically has multiple city gate stations serving different geographic portions of its service territory; the number and arrangement of city gates determine the minimum attack scale required to depressurize the full service territory.

2.2 District Regulator Stations

District regulator stations reduce gas pressure from the intermediate distribution pressure at which gas enters a neighborhood to the lower pressure at which residential and commercial customers receive service. They are the pressure control nodes within the distribution system, and their configuration determines how a pressure loss at one point in the system propagates to adjacent areas. District regulators are typically unmanned, automated, and numerous — a major urban LDC may operate hundreds of district regulator stations across its service territory. They are individually low-consequence targets; their significance in the DBA-MA analysis is as the mechanism by which a SCADA-manipulated under-pressure propagates from a single injection point across a distribution district.

2.3 Distribution Mains and Service Lines

Distribution mains are the intermediate-pressure pipelines that carry gas from city gate and district regulator stations through neighborhoods and streets to the service lines that connect individual customers. Service lines connect distribution mains to the individual customer meter at the building exterior. The distribution main and service line network is geographically extensive — a major urban LDC may operate tens of thousands of miles of distribution pipeline — and physically buried, making it inherently resistant to surface-level kinetic attack. The primary vulnerability of the mains and service line network is not physical destruction but pressure management: an adversary who controls the pressure at the district regulator level controls the pressure at every customer meter downstream.

2.4 LDC Peaking Resources

LNG peaking facilities. Many LDCs operate small-scale LNG liquefaction and storage facilities that supplement transmission supply during peak demand periods. LNG peaking facilities liquify gas during summer low-demand periods and vaporize it during winter demand peaks when transmission system deliveries are insufficient. LNG peaking storage provides the LDC's primary emergency supply resource for short-duration transmission disruptions; its capacity is typically sized for days to weeks of peak-day demand supplement, not a full-season supply replacement. LNG peaking facilities are within the scope of DBA-GAS-FC4 as integral LDC infrastructure.

Propane-air systems. Some LDCs in areas with limited transmission system access operate propane-air systems — installations that mix propane with air to produce a gas mixture with similar combustion characteristics to natural gas — as an emergency backup supply. Propane-air systems are typically small-scale and serve limited geographic areas. Still, they are the most readily available emergency alternative supply for rural and small-community distribution systems whose transmission supply is disrupted.

2.5 Distribution SCADA and Operational Technology

Distribution SCADA systems monitor pressure and flow at city gate stations, district regulator stations, and key distribution system nodes, and provide remote supervisory control of pressure regulators, isolation valves, and odorization systems. Distribution SCADA is typically less sophisticated than transmission pipeline SCADA — many LDCs still operate significant portions of their distribution systems on scheduled manual patrol rather than continuous remote monitoring — but is increasingly automated as LDCs deploy advanced metering infrastructure (AMI) and smart distribution technologies.

The distribution SCADA environment is characterized by a greater diversity of equipment vintages, communications protocols, and vendor platforms than transmission SCADA, reflecting decades of incremental investment by utilities operating under state PUC cost recovery frameworks. This diversity creates both cybersecurity challenges — heterogeneous systems are harder to defend uniformly — and resilience advantages — a compromise of one vendor's platform does not automatically affect systems running on different platforms in the same service territory.

2.6 Regulatory Authority

PHMSA. 49 CFR Part 192 establishes safety standards for distribution pipeline systems, including leak survey requirements, damage prevention programs, emergency response planning, and operator qualification. PHMSA's distribution integrity management program (DIMP) requires LDCs to assess and mitigate risks to their distribution systems. PHMSA does not regulate distribution system security; its jurisdiction is physical safety. The Merrimack Valley over-pressurization event led to enhanced PHMSA oversight of distribution system pressure management, but did not produce mandatory cybersecurity requirements for distribution SCADA.

State Public Utility Commissions. LDCs are regulated by state PUCs for service quality, rates, and safety. State PUC oversight of distribution system security varies enormously across jurisdictions — some states have adopted security requirements modeled on federal pipeline security guidelines. In contrast, others have no distribution-specific security requirements. The state PUC regulatory patchwork is the governing regulatory gap for FC4. There is no national minimum security standard for local gas distribution systems, and the variation across state jurisdictions means that an adversary can select attack targets in the states with the least demanding security requirements.

TSA. TSA Pipeline Security Directives apply to pipeline owners and operators, and their applicability to local distribution companies — which are not pipeline operators in the interstate commerce sense — is less clearly established than for transmission operators. The absence of a clear TSA mandate for SCADA cybersecurity distribution is a gap that parallels the processing plant cybersecurity gap identified in DBA-GAS-FC2.

3. Design Basis: Local Distribution Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the local distribution facility class inherits the infrastructure failure event, operational event, and natural hazard event categories from the

DBA-GAS Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

City gate station failure and distribution system depressurization. The governing non-adversarial bounding event is a city gate station failure that eliminates gas supply to a major distribution district, allowing system pressure to decline to zero as gas continues to be consumed downstream. The Merrimack Valley event (2018) is the reference event for the consequences of pressure loss followed by over-pressurization, rather than loss alone — a regulator failure caused a high-pressure gas stream to enter a low-pressure distribution system, resulting in over-pressurization, gas releases, and fires and explosions at eighty-five structures across three communities. The inverse failure — full depressurization following city gate loss — produces the re-pressurization constraint as its governing recovery consequence rather than the immediate fire and explosion hazard of over-pressurization. Still, both are city gate failure modes, and both must be addressed in the non-adversarial Design Basis.

The main rupture in a congested urban environment. A major distribution main rupture in a dense urban area, resulting in a significant gas release in a confined below-grade environment — a street, a subway tunnel, a utility corridor — with potential for ignition and explosion. The bounding case involves a high-pressure intermediate distribution main at a major street intersection, where the gas release and potential explosion affect not only the distribution infrastructure but also the adjacent utility infrastructure, street-level structures, and subway or underground transportation systems. The cross-sector consequence — distribution gas release damaging electric infrastructure, telecommunications, and transportation in a shared underground utility corridor — is a governing Tier 3 consequence for this bounding event.

Distribution SCADA outage. Loss of the distribution SCADA system governing a major LDC service territory, requiring transition to manual field operations for pressure monitoring and control. Manual distribution system operation relies on field patrol and manual pressure measurement at district regulator stations, significantly degrading the LDC's ability to detect and respond to pressure anomalies or leak events. The bounding case involves a SCADA outage during a demand surge — a cold snap or heat wave — when field crews are already deployed on emergency response and are unavailable for manual monitoring duties.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the local distribution facility class is a coordinated attack against the city gate stations serving a major metropolitan distribution system, combined with SCADA manipulation that suppresses low-pressure alarms and delays the operator's recognition that the distribution system is depressurizing, executed during peak winter heating demand to maximize the life-safety consequence of the resulting supply outage and to impose the full re-pressurization constraint on the recovery.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with distribution SCADA pre-positioning capability and precision kinetic delivery. The distribution system's relative lack of mandatory cybersecurity standards and the diversity of its SCADA environment may make distribution SCADA access achievable at a lower operational threshold than transmission SCADA access. Still, the kinetic component requires detailed knowledge of city gate station locations and configurations that are available in PHMSA public data and state PUC filings.

Modality: Cyber-kinetic combined arms. Distribution SCADA compromise suppresses low-pressure alarms at district regulator stations and at the city gate, delaying operator recognition of the depressurization while customers continue to consume gas. Kinetic action against the primary city gate stations destroys the physical supply inlet simultaneously or shortly after the alarm suppression begins.

Target selection: The minimum set of city gate stations whose simultaneous loss depressurizes the largest portion of the service territory. For a major metropolitan LDC with three to six city gate stations, simultaneous destruction of two or three strategically selected stations — those that feed the densest residential areas or that are hydraulically non-redundant — depressurizes the majority of the service territory.

Timing: Peak winter heating demand — a multi-day cold snap forecast at least a week in advance, giving the adversary time to execute the SCADA pre-positioning while the transmission system is under maximum demand stress and the LDC's operational resources are already committed to demand response. The cold-snap timing maximizes the life-safety consequence of the resulting outage and minimizes the LDC's available response capacity.

Simultaneity: Coordinated attacks on two or more metropolitan LDCs in the same regional market within a 48-hour window, exhausting the mutual aid technician resources that a single-LDC event would draw upon from neighboring utilities. The mutual aid constraint is the distribution system analog of the specialty contractor constraint in the DBA-OIL and DBA-GAS upstream series: the re-pressurization workforce is finite, and simultaneous multi-LDC events extend every utility's recovery timeline.

3.3 Dual Design Basis Architecture

Consistent with the DBA-GAS Sector Framework (§5.4) and DBA-EN-SF (§4.2), a local natural gas distribution system may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three distribution-specific AI governance interaction points are identified for the FC4 Design Basis:

Automatic pressure management and district regulator control. ML systems that automatically adjust district regulator set points based on downstream demand forecasts, ambient temperature, and transmission system inlet pressure are below the Bright Line when performing deterministic pressure management functions within their trained operating envelopes. They are formally testable and may operate autonomously for routine pressure adjustments. However, automatic pressure management is precisely the attack surface exploited in Scenario B: an adversary who has pre-positioned in the distribution SCADA system can modify district regulator set points systematically downward — each adjustment appearing within normal operating bounds — producing a progressive district-wide under-pressure that starves customer appliances without triggering the automatic low-pressure shutdowns that would alert the operator. The AI Governance Design Basis must establish that automatic pressure management set points are bounded by independent hardware-enforced limits that cannot be modified through SCADA software commands, that cumulative set point deviations from baseline seasonal profiles trigger human review rather than automatic adjustment, and that any set point recommendation that reduces distribution system pressure toward the low-pressure shutdown threshold requires Command Broker authorization regardless of its apparent operational justification.

Advanced metering infrastructure, ML, and demand forecasting. AMI systems that aggregate smart meter data and provide real-time demand forecasting and load profiling for distribution operations are ML systems operating below the Bright Line when performing signal aggregation and statistical analysis within their trained envelopes. The adversarial concern is manipulation of AMI data to produce false demand signals that cause the automatic pressure management system to reduce supply pressure — a second-order attack pathway that works through the optimization layer rather than directly through the pressure control layer. The Design Basis requirement is that demand forecasting ML systems receive AMI data through paths that are architecturally independent of the pressure control command path, and that anomalous aggregate demand signals — particularly demand readings that are inconsistent with ambient temperature and historical patterns — trigger human review before they influence pressure management set points.

Emergency response and crew dispatch Gen AI. Gen AI advisory tools for LDC emergency response operations — tools that recommend crew dispatch priorities, mutual aid activation timing, and re-pressurization sequencing during a distribution system emergency — are above the Bright Line and require Command Broker authorization. The distribution emergency context is the most time-critical application of the Command Broker requirement in the DBA-GAS series: a Gen AI tool whose crew dispatch recommendations are accepted without human review during a major distribution system emergency can misdirect the limited field workforce that is the sole constraint on the re-pressurization timeline. The Command Broker for emergency response Gen AI tools must be the qualified distribution operations supervisor with authority over field crew deployment. That authorization must be local and auditable — not delegable to an automated dispatch system that translates Gen AI recommendations into crew assignments without human review.

4. Design Basis Accident Scenarios

4.1 Scenario A: Coordinated Attack on City Gate Stations — Urban Distribution Depressurization

4.1.1 Initiating Event

Coordinated kinetic attack against three of the five city gate stations serving a major metropolitan LDC with approximately 800,000 residential and commercial customers, executed on the second night of a forecast multi-day cold snap when outdoor temperatures are below 10°F and distribution system demand is at or near annual peak. The cyber component, having achieved distribution SCADA access over the preceding weeks, has suppressed the low-pressure alarm thresholds at the district regulator stations downstream of the targeted city gates, ensuring that the depressurization develops for 30 to 60 minutes before the operator detects it through means other than the SCADA alarm system.

The kinetic component executes at 2:00 AM when field crew availability is at its lowest: UAS-delivered charges against the regulator skids and odorizer systems at three strategically selected city gate stations — the two that feed the highest-density residential districts and the one that is hydraulically non-redundant for the downtown commercial and hospital district. The remaining two city gate stations, serving lower-density suburban areas, are left intact.

4.1.2 Progression Sequence

T+0: Three city gate stations were destroyed simultaneously. Distribution SCADA alarm suppression is active. Customers in the three affected districts continue consuming gas, drawing down system pressure. Customers in the two unaffected suburban districts continue receiving normal service.

T+0 to T+30 minutes: Distribution system pressure in the three affected districts is declining. SCADA alarm suppression prevents automated notification. Field patrol is not yet detecting the pressure loss. Customers begin experiencing reduced flame at appliances — the first detectable symptom of low distribution pressure — but at 2:00 AM, most residential customers are asleep.

T+30 to T+60 minutes: Pressure in the highest-demand residential districts approaches low-pressure shutdown thresholds. Some automatic safety shutoffs at customer meters actuate — gas service to those customers is automatically interrupted, but the customer receives no notification. LDC control room operator detects the anomaly through a combination of demand readings inconsistent with expected consumption and SCADA data showing declining pressure at some nodes not covered by the alarm suppression. Emergency response activated.

T+60 minutes to T+6 hours: Full extent of the city gate destruction confirmed by field crews responding to the affected stations. Distribution system pressure in the three affected districts continues to decline as residual gas inventory in the mains is consumed. By approximately T+4 hours, the affected districts have reached zero pressure. Automatic safety shutoffs at customer meters have been activated across thousands of locations; many customers' appliances are now in an unsafe state with no gas flow. Emergency shut-in of the full distribution system in the affected districts was initiated to prevent gas odor complaints and to prepare for controlled re-pressurization. LDC activates mutual aid requests to neighboring utilities.

T+6 hours to T+72 hours: Three districts representing approximately 600,000 customers are at zero pressure in sub-10°F outdoor temperatures. The LDC's LNG peaking facility provides supplemental supply to the two unaffected districts, maintaining service there. Emergency coordination with the city and state emergency management agencies. Warming center activations across the three affected districts: hospitals and other critical facilities operating on backup heating systems. Mutual aid crews are beginning to arrive from neighboring utilities — mobilization requires 24 to 48 hours for crews from distant states.

T+72 hours onward: Re-pressurization operations beginning in the affected districts using the two remaining city gate stations, which can supply the full service territory at reduced pressure through interconnections between district zones. However, before any customer can receive gas, their service connection must be visited by a technician. With approximately 600,000 customers across three districts, re-pressurization at a rate of 500 to 800 customer visits per day per field crew — requiring all available LDC and mutual aid crews — produces a re-pressurization timeline of 3 to 5 weeks to restore service to all customers.

4.1.3 Consequence Envelope

Life-safety consequence. Three districts of a major metropolitan area at zero gas pressure in sub-10°F temperatures for 3 to 5 weeks of progressive re-pressurization. Vulnerable residential populations — elderly, medically dependent, and economically isolated customers — are at direct risk of hypothermia-related injury and death during the period before their service is restored. The life-safety consequence is concentrated in the highest-density residential areas served first by the city gate stations that were destroyed. Emergency shelter capacity in the affected city is measured in hundreds or low thousands of places; the affected population is hundreds of thousands.

Re-pressurization is the governing constraint. The recovery timeline is not governed by the city gate station reconstruction timeline — emergency city gate stations can be deployed or the existing stations partially repaired within days to weeks. The workforce constraint governs it: there are not enough qualified gas technicians in the regional mutual aid system to visit 600,000 customer locations in less than 3 to 5 weeks. No investment in city gate station resilience, no amount of spare equipment pre-positioning, and no emergency regulatory action can reduce this timeline by more than a modest factor. The only durable mitigation is reducing the number of customers who lose pressure by preventing the depressurization in the first place, or by designing the distribution system with hydraulic zones that limit the geographic extent of any single city gate failure.

Simultaneous attack amplification. If a simultaneous Scenario A attack occurs at a second metropolitan LDC in the same regional mutual aid system within 48 hours, the mutual aid workforce available to the first LDC is divided between two emergencies. The re-pressurization timeline for each LDC extends from 3 to 5 weeks to 5 to 8 weeks or longer, compounding the life-safety consequence across both service territories.

4.2 Scenario B: SCADA-Induced Systematic Under-Pressure — The Slow Drain

4.2.1 Initiating Event

Cyber intrusion into the distribution SCADA system of a major LDC, with pre-positioned access to the automatic pressure management systems governing district regulator set points across a major urban distribution district. The adversary initiates a systematic downward adjustment of district regulator set points across a targeted district, not to zero pressure, but to a level 15 to 25 percent below the minimum required for reliable appliance operation. Each set point adjustment is within the normal operating range; the aggregate effect is a district-wide under-pressure condition that develops over 6 to 12 hours as demand cycles through the day.

4.2.2 The Gradual Consequence

Under-pressure in a distribution system produces consequences that are gradual, diffuse, and initially ambiguous. Customer appliances operate unreliably — burners that fail to light, furnace ignition cycles that fail to sustain flame, water heaters that produce inadequate hot water. Individual customers report problems to the utility's customer service line; each complaint appears to reflect an appliance problem rather than a supply problem. The aggregate pattern of complaints is not immediately visible to the distribution operations team, whose SCADA system is showing pressure values that, while below optimal, are within the range that can be attributed to high demand rather than a supply problem.

The slow drain exploits the boundary between a demand-driven pressure reduction — which operators expect and manage during cold weather — and an adversary-induced pressure reduction. Both look the same on the SCADA screen. The detection challenge is not identifying that pressure is low; it is identifying that the pressure is low for the wrong reason. By the time the pattern of customer complaints triggers a systematic investigation of the district regulator set points, the affected district may have been operating at degraded pressure for 12 to 24 hours.

4.2.3 Consequence Envelope: Systematic Under-Pressure

Diffuse supply degradation. Thousands of residential and commercial customers in the affected district are experiencing unreliable gas appliance operation for 12 to 24 hours before the problem is identified and remediated. The consequence is not a full supply outage — customers do not lose gas entirely — but a systematic degradation of service quality that produces customer discomfort and appliance cycling stress without the immediate life-safety urgency of a full depressurization.

Detection and attribution delay. The primary consequence of Scenario B is not the immediate supply degradation but the detection delay that allows the condition to persist and potentially compound. An adversary who has demonstrated the ability to manipulate district regulator set points through the distribution SCADA system without triggering automated alarms has demonstrated a capability that could be escalated to full depressurization in a subsequent operation. The Scenario B intrusion, if undetected, provides the adversary with validated SCADA access and tested alarm suppression capability that can be used for a subsequent Scenario A operation with higher kinetic consequence.

Appliance safety consequence. Extended under-pressure operation can cause gas appliance safety systems to cycle repeatedly — ignition attempts that fail due to insufficient gas pressure leave gas partially released at the burner before the safety system shuts off the valve. In poorly maintained appliances or in installations where the safety system function is degraded, repeated incomplete ignition cycles under low-pressure conditions can create gas accumulation risk. This is a secondary consequence that is difficult to quantify but must be acknowledged in the DBA-MA consequence envelope.

4.3 Scenario C: Cold-Weather Compound Attack — Transmission and Distribution Simultaneous Failure

4.3.1 Initiating Event

Coordinated attack that simultaneously executes a Scenario A distribution city gate attack against one or more metropolitan LDCs and a DBA-GAS-FC1 Scenario A transmission compressor station attack against the transmission corridor that supplies those LDCs, during a multi-day winter cold snap with outdoor temperatures forecast to remain below 15°F for at least 10 days. The transmission attack reduces inlet pressure at the LDCs' city gate stations before the kinetic component destroys the city gates themselves — the under-pressure at the city gate arrives before the gate is destroyed, providing a brief window during which the LDC's automatic pressure management systems are already compensating for low inlet pressure when the city gate destruction eliminates inlet supply.

4.3.2 The Compound Supply Failure

The Scenario C compound attack eliminates both the primary supply source — transmission pipeline supply at the city gate — and the secondary supply supplement — LDC LNG peaking facilities, which depend on vaporized LNG to supplement transmission supply when inlet pressure is low. The transmission attack that reduces inlet pressure at the city gate simultaneously reduces the pressure differential available to drive LNG peaking vaporization at the required delivery rate. The transmission attack partially degrades the LDC's emergency supply resources before the city gate destruction eliminates the primary supply.

The compound attack also eliminates the mutual aid supply path. During a single-LDC city gate outage, neighboring LDCs can sometimes supply emergency gas through distribution system interconnects — border valves that allow gas flow between adjacent service territories. When the transmission attack affects the entire corridor, all LDCs in the region face reduced supply simultaneously; there is no neighboring LDC with surplus gas to share through a distribution interconnect.

4.3.3 Consequence Envelope: Compound Attack

Regional residential heating failure. Multiple metropolitan LDCs simultaneously at zero or near-zero gas pressure during a 10-day cold snap, with no transmission supply available to restore service and no mutual aid gas supply path from adjacent service territories. The combination of full depressurization, cold temperatures, and simultaneous failure across multiple LDCs produces a regional residential heating emergency that exceeds the capacity of emergency shelter systems, emergency heating fuel distribution programs, and mutual aid workforce resources simultaneously. This is the most severe consequence envelope in the DBA-GAS series for direct residential life-safety impact.

Hospital and critical facility consequence. Hospitals, nursing homes, and other critical facilities in the affected service territories that use natural gas for heating, sterilization, and cooking — and that may not have backup heating systems sized for a multi-week gas outage — face operational continuity challenges that compound the direct patient care demands of cold-weather exposure casualties. The hospital consequence is a Tier 3 cross-sector cascade that requires DOD and FEMA coordination independent of the gas sector response.

Electric system interaction. The simultaneous loss of gas-fired generation in the transmission attack component (via the gas-electric cascade described in DBA-GAS-FC1 and DBA-ES-GC-FC4) and the increase in electric heating demand as affected customers switch to supplemental electric heating produce a compound stress on the electric system that is the winter Uri scenario with deliberate adversarial initiation. The compound gas-electric consequence is the governing Tier 3 cascade event for Scenario C.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in LDC distribution SCADA systems or city gate station infrastructure. Anomalous distribution SCADA network activity: unusual access to pressure management set point archives, anomalous vendor remote access sessions, and lateral movement inconsistent with normal maintenance operations. Patterns of city gate station physical surveillance across multiple LDC service territories in the same regional market. Procurement inquiries for city gate regulator specifications or distribution SCADA vendor platform documentation. Unusual aerial activity over city gate station locations. Aggregate distribution pressure readings are trending systematically below seasonal norms across multiple district regulator zones without a corresponding demand explanation.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized distribution of SCADA access. Systematic low-pressure readings across multiple district regulator zones are inconsistent with demand or inlet pressure conditions. Confirmed unauthorized UAS in the city gate station airspace. Simultaneous pressure anomalies at city gate stations from two or more LDCs in the same regional market within 24 hours. Confirmed kinetic event at any city gate station. Pattern of residential appliance complaints consistent with systematic under-pressure across a distribution district rather than isolated appliance failures.

5.2 Decision Authority Framework

LDC operator level: Distribution operations supervisor authority to initiate emergency shut-in of distribution districts, activate emergency LNG peaking supply, and request mutual aid workforce. PHMSA notification for distribution system incidents under emergency response plan requirements. State PUC notification. Local emergency management notification for supply disruptions affecting residential heating during cold weather. TSA and CISA notification for confirmed or suspected distribution of SCADA cyber incidents.

State coordination: State PUC emergency authority over LDC operations during declared energy emergencies. State emergency management coordination for warming center activation and vulnerable population response. Governor's office coordination for National Guard activation for welfare check operations in affected service territories.

Federal coordination: DOE ESF-12 for regional supply disruption response. CISA for the distribution of SCADA cyber incident coordination. FEMA for mass care and emergency shelter coordination when the life-safety consequence exceeds local and state response capacity. FBI and DOD for events with confirmed or strongly suspected state-actor attribution. HHS for medical consequence coordination when the hospital and nursing home gas supply is affected.

5.3 Federal Coordination Thresholds

Confirmed kinetic attack on any city gate station. Confirmed distribution SCADA compromise with suspected nation-state attribution. The distribution system depressurization is affecting more than 100,000 residential customers during the heating season, with simultaneous supply disruptions at two or more LDCs in the same regional market within 48 hours. Life-safety events — confirmed hypothermia casualties — attributable to residential heating gas supply loss. Compound gas-electric cascade conditions are developing due to simultaneous transmission and distribution disruptions.

6. Recovery Architecture

6.1 Re-Pressurization as the Governing Recovery Constraint

The re-pressurization constraint is the central organizing principle of FC4 recovery architecture and distinguishes it from every other recovery architecture in the DBA-GAS series. All other series recovery timelines are governed by equipment procurement, regulatory authorization, or physical repair — constraints that can, in principle, be accelerated by pre-positioning equipment, expediting procurement, or streamlining regulation. None of these mechanisms can accelerate the re-pressurization workforce constraint: it is a physical constraint on the number of qualified gas technicians available to visit customer locations, and it can only be reduced by reducing the number of customers who must be visited or by increasing the workforce available.

The two levers available to reduce the re-pressurization timeline are hydraulic zone design and mutual aid workforce pre-positioning. Hydraulic zone design — the configuration of isolation valves in the distribution system that limit how far a pressure loss propagates from a single city gate failure — determines how many customers are affected by a given attack. A distribution system with well-designed hydraulic zones may limit a two-city-gate attack to 200,000 customers rather than 600,000, cutting the re-pressurization workforce requirement by two-thirds. Mutual aid workforce pre-positioning — pre-established agreements with neighboring utilities and with specialized pipeline contractor firms for emergency workforce deployment — determines how quickly the available re-pressurization workforce can be assembled. Both require deliberate investment decisions; neither exists as a mandatory requirement in any current regulatory framework.

6.2 Distribution System Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency district shut-in to stop pressure loss and prevent gas odor complaints that could create explosion risks. City gate station damage assessment and emergency city gate procurement or temporary station deployment. LNG peaking facility activation to the maximum vaporization rate for unaffected districts. Mutual aid workforce activation — immediate notification of all pre-established mutual aid partners; workforce mobilization requires 24 to 48 hours minimum for distant utilities. Emergency management coordination for warming center activation and vulnerable population identification. State PUC and PHMSA notification. CISA notification for SCADA cyber component.

Phase 2 — Short-term (72 hours to 2 weeks). Emergency city gate station deployment or partial repair to restore supply to the highest-priority districts — hospitals, nursing homes, densely populated residential areas with the highest vulnerable population concentration. Re-pressurization operations begin in priority districts with all available field crews. Customer notification and door-hanger campaign identifying re-pressurization visit schedule. Warming center operations are sustaining vulnerable populations displaced from unheated residences. Mutual aid crews from distant utilities are arriving and being integrated into field operations.

Phase 3 — Medium-term (2 weeks to recovery). Progressive re-pressurization extending through all affected districts. Permanent city gate station repair or replacement procurement. SCADA forensic investigation and reconstitution — the SCADA system must be re-validated against its pressure management set point baseline before automated operations resume. Customer service surges to handle re-lighting requests and appliance safety complaints. State PUC reporting and incident investigation.

6.3 Critical Recovery Constraints

Re-pressurization workforce. The governing constraint for Scenario A and C events. The available pool of qualified gas technicians in a regional mutual aid system is finite. For a major metropolitan LDC with 600,000 affected customers, the re-pressurization timeline under maximum mutual aid deployment is 3 to 5 weeks. For a simultaneous two-LDC event, each LDC's timeline extends to 5 to 8 weeks or more. Pre-established mutual aid agreements with neighboring utilities — including agreements with non-adjacent utilities in other regions for distant mutual aid — are the primary mitigation, and their existence and adequacy should be verified against the Scenario A and C bounding events, not against the single-city-gate failure scenarios for which most mutual aid agreements were designed.

Emergency city gate station deployment. Temporary or mobile city gate stations — pre-engineered modular pressure regulation skids that can be connected to the transmission system at a new location — are commercially available but not widely pre-positioned. Deployment requires a suitable connection point on the transmission system, regulatory authorization, and installation by qualified pipeline contractors. Under emergency conditions, a temporary city gate station can be operational in 3 to 10 days, depending on the connection complexity. Pre-positioned temporary city gate equipment at strategic locations in major metropolitan service territories would reduce the window during which the affected districts must operate at zero pressure.

SCADA reconstitution and set point re-validation. For Scenario B and the cyber components of Scenario A and C, distribution SCADA forensic investigation and reconstitution must be completed before automated pressure management can resume. The minimum 4 to 6 week timeline is consistent with other DBA-GAS series instruments. The distribution context adds the requirement that pressure management set point baselines be independently verified against historical pressure logs before automated set point adjustment is re-enabled — the reconstituted SCADA system must be re-certified to operate within the pressure management envelope established by the Design Basis.

Vulnerable population welfare check capacity. During a major urban distribution outage in winter, the welfare check capacity of local emergency services — police, fire, and social service agencies responsible for contacting vulnerable residents without heat — is a binding constraint on life-safety consequence management. The number of vulnerable residents who cannot self-evacuate to warming centers in a major metropolitan area during a winter distribution outage can number in the tens of thousands; local emergency services cannot contact all of them in the time available before cold-related health consequences develop. National Guard activation for welfare check operations is the federal response mechanism, but activation and deployment require days.

7. Gaps and Recommendations

7.1 Regulatory Gap: Absence of National Minimum Security Standards for Distribution Systems

Local gas distribution systems — the infrastructure that directly serves the residential population — have no national minimum security standard. PHMSA regulates distribution pipeline safety; state PUCs regulate service quality and rates; TSA's applicability to LDCs is unclear; and no federal agency has issued mandatory cybersecurity requirements for distribution SCADA systems. The regulatory patchwork across fifty state PUC jurisdictions produces enormous variation in distribution system security posture, with some states having no distribution-specific security requirements at all. A national minimum security standard for local gas distribution systems — covering both physical security of city gate stations and cybersecurity of distribution SCADA — should be developed by TSA in coordination with PHMSA, with a risk-tiered framework that applies the most stringent requirements to LDCs serving the largest populations in the coldest climates.

7.2 Hydraulic Zone Design Standard

The re-pressurization workforce constraint can only be materially reduced by limiting the geographic extent of pressure loss from a single city gate failure, which is a function of the distribution system's hydraulic zone design. No mandatory standard requires LDCs to design their distribution systems with hydraulic isolation capability sufficient to limit a single city gate failure to a defined maximum number of customers. PHMSA's distribution integrity management program requires risk assessment but does not specify hydraulic zone design outcomes. A performance-based hydraulic zone design standard — requiring that any single city gate failure affect no more than a defined maximum customer count before isolation is achievable — would be the highest-consequence single mitigation for the Scenario A re-pressurization constraint. PHMSA should develop this standard as part of the next DIMP revision.

7.3 Mutual Aid Workforce Pre-Positioning

Pre-established mutual aid agreements for distribution emergency response workforce exist among some neighboring LDCs, but are not universally in place, are not sized for the Scenario A or C bounding events, and do not address the distant-utility mutual aid required for a simultaneous multi-LDC event. The American Gas Association (AGA) mutual aid framework provides the voluntary foundation; PHMSA should assess whether mandatory minimum mutual aid requirements — specifying the minimum pre-established workforce commitments and activation timelines for LDCs above a defined customer count threshold — should be added to the distribution emergency response plan requirements in 49 CFR Part 192. The assessment should use the Scenario A re-pressurization workforce requirement as the design basis for minimum mutual aid sizing.

7.4 Emergency City Gate Station Pre-Positioning

No federal or industry program pre-positions temporary or mobile city gate stations at locations that would allow rapid deployment to replace a destroyed city gate in a major metropolitan service territory. The electric sector's large power transformer strategic reserve program provides the analytical model. DOE, in coordination with PHMSA and the AGA, should assess the feasibility of a pre-positioned temporary city gate station program for the highest-consequence metropolitan LDC service territories — those serving the largest populations in the coldest climates with the fewest hydraulic redundancy options. The program should identify the connection points on the transmission system where temporary stations could be deployed, the regulatory authorization pathway for emergency deployment, and the equipment specifications required.

7.5 Vulnerable Population Life-Safety Emergency Response Framework

No federal framework specifically addresses the life-safety consequence management for a major urban gas distribution outage in winter — the combination of direct cold exposure risk, displacement of hundreds of thousands of residents from unheated homes, and demand surge on emergency shelter and medical systems that a Scenario A or C event would produce. FEMA's mass care emergency support function addresses displaced populations but has not been specifically calibrated for the scale and duration of a major urban gas outage. DOE, FEMA, and HHS should jointly develop a gas distribution emergency life-safety response framework that defines the federal coordination mechanisms, resource pre-positioning requirements, and interagency decision authorities for a metropolitan distribution outage affecting more than 100,000 residential customers during heating season. The Scenario A consequence envelope should be the design basis for this framework.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the local natural gas distribution facility class. Issued under DBA-GAS Sector Framework v1.5. Three scenarios were developed: Scenario A (coordinated city gate station attack — urban distribution depressurization with re-pressurization workforce as governing recovery constraint), Scenario B (SCADA-induced systematic under-pressure — the slow drain, exploiting automatic pressure management manipulation without physical attack), Scenario C (cold-weather compound attack — simultaneous transmission and distribution failure producing regional residential heating emergency). Dual Design Basis architecture established with three distribution-specific AI governance interaction points. The re-pressurization constraint is established as the governing recovery metric unique to this facility class. Cross-sector consequence XC-4 (residential heating dependency) from DBA-ES-GC-FC4 addressed from the gas sector perspective, resolving DBA-EN-SF OI-4. Regulatory gaps identified: national minimum distribution security standard, hydraulic zone design standard, mutual aid workforce pre-positioning, emergency city gate station pre-positioning, and vulnerable population life-safety response framework. Completes the initial DBA-GAS facility class instrument set: FC1 through FC4, all at v0.1 or above.